Exercise 1: SQL Fundamentals

1. SELECT *
FROM employees;

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55 000	2018-06-15	New York
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52 000	2016-04-10	San Francisco
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
8	Jessica	Moore	HR	51 000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	~~HR~~ IT	50 000	2020-08-10	San Francisco

2. SELECT DISTINCT (department)
From employees;

department
IT
HR
Finance
Marketing

3.
```
SELECT first_name,
       last_name,
       salary
FROM employees
ORDER BY salary DESC;
```

first_name	last_name	salary
~~Mike~~	~~Johnson~~	~~60000~~
Emily	Davis	62000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55000
Daniel	Clark	53 000
Sarah	Brown	53 000
David	White	52 000
Jessica	Moore	51 000
Laura	Hall	50 000
Jane	Smith	48 000

4.
```
SELECT id,
       salary
FROM employees
ORDER BY salary DESC
Limit 5;
```

id	salary
6	62000
3	60 000
7	59 000
1	55 000
4	53 000

5
```
SELECT last_name, department
FROM employees
WHERE department='IT';
```

last_name	department
Doe	IT
Brown	IT
Davis	IT
Hall	IT

6.
```
SELECT last_name, department, salary
FROM employees
WHERE department='Finance' AND salary > 58 000;
```

last_name	department	salary
Johnson	Finance	60 000
Wilson	Finance	59 000

7 SELECT last_name,
department
From employees
WHERE department='HR' OR department='Marketing';

last_name	department
Smith	HR
White	Marketing
Moore	HR
Clark	Marketing

8 SELECT last_name,
department,
FROM employees
WHERE department='IT';

last_name	department
Smith	HR
Johnson	Finance
White	Marketing
Wilson	Finance
Moore	HR
Clark	Marketing

9 SELECT last_name,
department
FROM employees
WHERE department IN ('HR','IT','Finance');

last_name	department
Doe	IT
Smith	HR
Johnson	Finance
Brown	IT
Davis	IT
Wilson	Finance
Moore	HR
Hall	IT

10 SELECT last_name,
department,
salary
FROM employees
WHERE department='IT' AND salary>50000 AND city='New York';

last_name	department	salary
Doe	IT	55000
Brown	IT	53 000

11 SELECT first_name,
last_name,
department,
salary
FROM employees
WHERE department='Finance' OR department='Marketing' AND salary>52000
ORDER BY salary DESC;

first_name	last_name	department	salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000
Daniel	Clark	Marketing	53 000

12 SELECT DISTINCT city,
department
FROM employees
WHERE department IN('Finance','Marketing');

city	department
Los Angeles	Finance
San Francisco	Marketing
Houston	Finance
Chicago	Marketing

13 SELECT last_name,
department,
salary
FROM employees
WHERE department != 'Finance' AND salary >50 000
ORDER BY hire_date ASC;

last_name	department	salary
Davis	HR	51 000
White	Marketing	52 000
Moore	HR	51 000
Doe	IT	55 000
Brown	IT	53 000

14 SELECT last_name,
department,
city
FROM employees
WHERE ~~department IN('Chicago', Los Angeles') AND~~ OR
~~city department.city IN('Chicago', 'Los Angeles') AND~~
department = 'IT' AND department = 'Marketing'
LIMIT 3;

last_name	department	city
Davis	IT	Chicago
Clark	Marketing	Chicago

14 SELECT last_name,
department,
city
FROM employees
WHERE department IN('IT', 'Marketing') AND
city IN('Chicago', Los Angeles')
LIMIT 3;